[Properties & formula]

Matrix

→ $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}_{m \times n}$ ← Row

↑ ↑ ↑

column

Types of matrix :- [Important types] -

① zero or null matrix - $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

② Diagonal matrix - $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

③ Unit or Identity matrix-

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

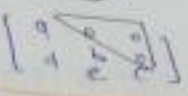
→ Transpose of a matrix-

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$A' = A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

[Interchange of row and column]

Triangular matrix → Upper →

lower → 

Properties of transpose of matrix-

1. $(A')' = A$
2. $(KA)' = KA'$ [K = constant]
3. $(A+B)' = A' + B'$
4. $(AB)' = B'A'$ ⟶ very important.

Symmetric matrix-

$$A = A'$$

$$A = \begin{bmatrix} 4 & 3 \\ 3 & 5 \end{bmatrix}, \quad A' = \begin{bmatrix} 4 & 3 \\ 3 & 5 \end{bmatrix} \Rightarrow \boxed{A = A'}$$

~~Skey~~ ~~Sk~~ Skew-symmetric matrix- [Diagonal element always 0]

$$A' = -A$$

$$A = \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}, \quad A' = \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix} \Rightarrow \boxed{A' = -A}$$

Properties of symmetric and skey-symmetric matrix :-

1. Sum of two symmetric matrix is a symmetric matrix.
2. If A is symmetric matrix then, KA is also symmetric matrix.
3. Sum of two skew-symmetric matrix is skew-symmetric.
4. If A is skew-symmetric matrix then, KA is also skew symmetric.
5. • $A + A'$ = symmetric matrix (always)
   • $A - A'$ = skew-symmetric matrix (always)

Determinant

$$\det A = |A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$|A| = (a_{11} \cdot a_{22}) - (a_{12} \times a_{21})$$

$$|A| = \begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix} = (2 \times 2) - (4 \times 1)$$
$$= 0$$

→ Properties of Determinant -

① $|A| = |A^T|$

② if two rows or clouma of a determinant are identical (or same) then determinant = 0.

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 5 & 4 & 11 \\ 1 & 2 & 3 \end{vmatrix} = 0 \quad [R_1 = R_2]$$

③ $|KA| = k^n . |A|$ [K = constant]
[n = order of matrix]

Eg.- $|A| = 7$ of order (3×3)

$$|3A| = 3^n |A|$$
$$= 3^3 |A|$$
$$= 27 \times 7$$
$$= \underline{189}$$

- Minor → deletion of row and column. (M_{ij})

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$M_{11} = 1 = 4$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$M_{12} = \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = (4\times9) - (7\times6)$$
$$= 36 - 42$$
$$M_{12} = -6$$

- Cofactor → $C_{ij} = (-1)^{i+j} \cdot M_{ij}$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$C_{11} = (-1)^{1+1} \cdot 4$
$= 4$

$C_{12} = (-1)^{1+2} \cdot 3$
$= -3$

$C_{21} = (-1)^{2+1} \cdot 2$
$= -2$

$C_{22} = (-1)^{2+2} \cdot 1$
$= 1$

~~C_{ij} of~~ cofactor of $A = \begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix}$

→ Adjoint → $adj(A) = (\text{cofactor})^T$
= transpose of cofactor

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

cofactor of A = $\begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix}$

$$\begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix}^T = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$adj(A) = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

→ Inverse of a matrix = (A^{-1})

$$A^{-1} = \frac{adj(A)}{|A|}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$|A| = (1 \times 4) - (2 \times 3)$$
$$= -2$$

$$adj(A) = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}}{-2} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$

Properties of Inverse of a matrix:-

1. $(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$
2. $(A^T)^{-1} = (A^{-1})^T$
3. $|adj(A)| = |A|^{n-1}$
4. $adj(adj(A)) = |A|^{n-2} \cdot A$.
5. $A \cdot A^{-1} = I$ ------ → very important
6. $|A^{-1}| = \frac{1}{|A|}$ or $|KA^{-1}| = \frac{1}{|KA|} = \frac{1}{|K^n|A|}$

[n = order of matrix]

Singular matrix-

$$|A| = 0$$

$$|A| = \begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix} = 0$$

Non-Singular matrix-

$$|A| \neq 0$$

$$|A| = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2$$

$$|A| \neq 0$$

→ Rank of a Matrix :- $\rho(A)$ - Non-zero rows.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 12 \end{bmatrix}$$

applying $R_2 \to R_2 - 2R_1$ & $R_3 \to R_3 - 3R_1$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of matrix $A = \rho(A) = 1$

→ Properties :-

1. $\rho(A) = \rho(A^T)$
2. Rank of a non-singular square matrix of order n is n.
3. If a matrix has a non-zero minor of order r, its rank is $\geq r$.
   if all minors of a matrix of order $r+1$ are zero, its rank is $\leq r$.

→ Gauss Jordan method to find inverse :-

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 3 & 2 & 4 & 0 & 1 & 0 \\ 5 & 1 & 6 & 0 & 0 & 1 \end{array}\right]$$

Transform into I — Transform (automatically) into Inverse

Note:- In gauss-jordan method; Only Row-elementary operation is applicable not column operation.

Normal form →

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

r = rank

Eg:- $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 1 & 0 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$.

rank = 3 ✓

PAQ form →

$$A = P\,A\,Q$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

↓ Change into Normal form

↓ only row elementary operation

↓ only column elementary operation.

→ Solution of linear system of equations,

① Method of determinants – Cramer's rule :-

$$\begin{bmatrix} a_1x+b_1y+c_1z=d_1 \\ a_2x+b_2y+c_2z=d_2 \\ a_3x+b_3y+c_3z=d_3 \end{bmatrix}$$

find $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$$x = \frac{1}{\Delta}\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

$$y = \frac{1}{\Delta}\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$z = \frac{1}{\Delta}\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

Matrix inversion method -

$$\begin{bmatrix} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{bmatrix}$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad D = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$AX = D$$
$$X = A^{-1}D$$

→ System of linear Equation:-

① Non-Homogenous:- $A \cdot X = D$

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

* $A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$

$$K = \begin{bmatrix} a_1 & b_1 & : d_1 \\ a_2 & b_2 & : d_2 \end{bmatrix}$$

① for unique Solution :- [Consistent]

Rank of A = Rank of K = n (no. of unknown (x, y, z etc))

② for infinety many solution :- [consistent]

Rank of A = Rank of K < n (no. of unknown (x, y, z, etc))

③ for no solution :- [Inconsistent]

Rank of A ≠ Rank of K

Note :- Summary

Consistent
↓
$\rho(A) = \rho(K)$

$\rho(A) = \rho(K) = n$
↓
Unique soln

$\rho(A) = \rho(K) < n$
↓
Infinetly many soln

Inconsistent
↓
$\rho(A) \neq \rho(K)$
↓
No soln

Note :- To find unique solution → easy
To find infinetly many soln → take parameter [n-r] (k_1, k_2, etc)
To find No soln → No soln [$x = y = z = 0$]

② Homogenous Solution:-

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\therefore$ * $A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$

① for trivial soln [zero soln]:- [consistent]

Rank of $A = n$ (no. of unknowns (x, y, z etc).

$x = y = z = 0$

② for Non-trivial soln [infinitely many soln]:- [consistent]

Rank of $A \neq n$ (no. of unknowns (x, y, z etc).

* $[\,|A| = 0\,]$

→ Orthogonal matrix:-

$$A^{-1} = A'$$

properties:-

① $(A')' = (A^{-1})' = (A')^{-1} = A$

② If A is orthogonal then,

$$|A| = \pm 1$$

Now;

$$A \cdot A^{-1} = I$$

for checking of orthogonal

$$\boxed{A \cdot A' = I}$$

→ Orthogonal transformation:-

given equation → $Y_1 = a_1 x + b_1 y$

$Y_2 = a_2 x + b_2 y$

$$Y = AX$$

$$X = A^{-1} Y$$

- To check "regular" ⇒ matrix is non-singular
  $|A| \neq 0$

- Inverse transformation ⇒ $X = A^{-1} Y$ → change into equation.

→ Eigen Values :—

$$|A - \lambda I| = 0$$

value of $\lambda = \ldots$ ⟶ Eigen values.

→ Eigen Vector :—

$$[A - \lambda I][X] = 0$$

checking at eigen values ($\lambda = \ldots$)

$$\frac{x}{a_1} = \frac{y}{a_2} = \frac{z}{a_3} = k$$

Note : If more equation then take two equation ~~that~~ give one value

$k(a_1, a_2, a_3)$ ⟶ Eigen vector.

→ Properties :—

① Any square matrix A, its transpose A' have the same eigen values.

② The eigen values of a triangular matrix are just the just diagonal element of the matrix.

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ 0 & a_4 & a_5 \\ 0 & 0 & a_6 \end{bmatrix}$$

→ Eigen value $= a_1, a_4, a_6$

→ Eigen vector if $a_6 = 0$

$\frac{a_3}{\cancel{3}} = \cancel{k}$.

③ The eigen values of an idempotent matrix are either zero or unity.

idempotent → $A = A^2$

$\lambda = 0$ or 1

④ The sum of the eigen value of a matrix is its trace i.e. the sum of the element of the principal diagonal.

⑤ The product of the eigen value of a matrix A is equal to its determinant.

⑥ If λ is an eigen value of a matrix A, then $\frac{1}{\lambda}$ is the eigen value of A^{-1}.

⑦ If λ is an eigen value of an orthogonal matrix, then $\frac{1}{\lambda}$ is also eigen value.

⑧ If $\lambda_1, \lambda_2, \lambda_3, \ldots$ are the eigen values of a matrix A, then A^m has the eigen value $\lambda_1^m, \lambda_2^m, \lambda_3^m \ldots$ $[m > 0]$

Cayley- Hamilton Theorem:-

$$|A - \lambda I| = 0$$

$$a\lambda^n + b\lambda^{n-1} + c\lambda^{n-2} \cdots + d = 0$$

by applying ~~Characterstic~~ Cayley hamilton theorem

$$aA^n + bA^{n-1} + cA^{n-2} \cdots dI = 0$$

this equation is known as characterstic equation

If this equation verify i.e its equal to 0 then its verify cayley hamilton theorem.

(*) We can find Inverse Using this characterstic eqn.

Eg: $A^2 - 4A - 5I = 0$

multiply this eq- by A^{-1}

$$A^2 \cdot A^{-1} - 4A \cdot A^{-1} - 5IA^{-1} = 0$$

$$A - 4I - 5A^{-1} = 0$$

$$A^{-1} = \frac{[A - 4I]}{5}$$

→ Reduction to diagonal form :-

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Step 1:- find eigen value, then eigen vector.

Step 2:- Write the eigen vector in column form.
This transformation called modal matrix also called 'P'.

Step 3:- The diagonal matrix has the eigen values of A as its diagonal elements.
Hence, Reduce to diagonal form;

$$\underset{\substack{\downarrow \\ \text{modal} \\ \text{matrix}}}{P^{-1}} \cdot A \cdot P = D$$

$$\text{Diagonal matrix} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_2 \end{bmatrix}$$

$[\lambda_1, \lambda_2, \lambda_3] \rightarrow$ Eigen value

$$P = \begin{bmatrix} u_1 & \omega_1 & \phi_1 \\ u_2 & \omega_2 & \phi_2 \\ u_3 & \omega_3 & \phi_3 \end{bmatrix}$$

- $u, \omega, \phi \rightarrow$ Eigen vectors of each eigen values

* Power of a matrix :-

$$D = P^{-1} \cdot A P \Rightarrow D^2 = P^{-1} A P \cdot P^{-1} A P$$
$$\Rightarrow D^2 = P^{-1} A^2 P$$

$$D^n = P^{-1} \cdot A^n \cdot P$$

$$\boxed{A^n = P \cdot D^n \cdot P^{-1}}$$

Step :- 1 - find Eigen value, then eigen vectors

Step :- 2 - find the corresponding modal matrix P (using eigen vector).

Step :- 3 - find diagonal matrix D from $D = P^{-1} A P$.

Step 4 :- Obtain A^n from

$$A^n = P \cdot D^n P^{-1}$$

Note :- Don't think $P \cdot D P^{-1}$

$$P P^{-1} = I$$

If $P P^{-1} = I$ but not in $P X P^{-1}$

because $P X \neq X P$

(*) To check ~~diagonalization~~ a matrix is diagonalizable or not :-

→ If matrix have same eigen vector then diagonalization not possible.

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 = 0$$

$$\lambda = 1 \quad (2 \text{ times})$$

Eigen vector at $\lambda = 1$

$$[A - \lambda I] \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$y = 0$$

$$\frac{x}{1} = \frac{y}{0}$$

Eigen vector $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$|P| = 0$

P^{-1} not possible $\therefore$ Not diagonalization.

- Similarity of matrices:-

A and B are similar if and only if

$$\boxed{B = P^{-1} A P}$$

This is called similarity transformation.

Axos